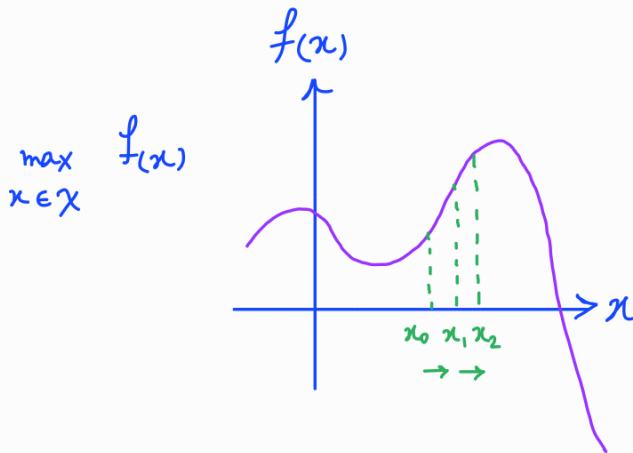
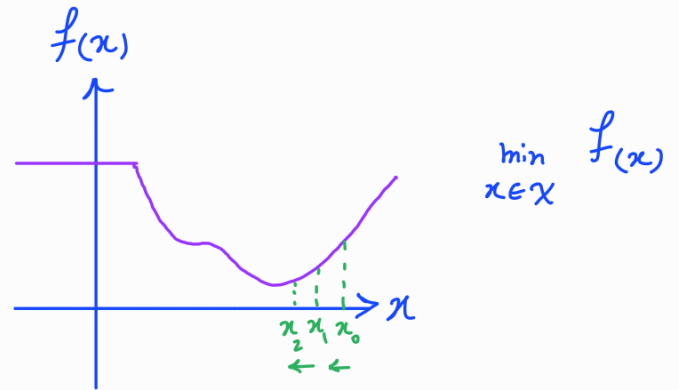


Gradient descent (ascent) and stochastic gradient descent (ascent):

A family of algorithms that aim to optimize, minimize or maximize, a function iteratively by moving toward the direction changing, decreasing or increasing, the function (maximally).



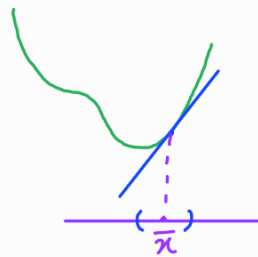
Gradient Ascent



Gradient Descent

$f: \mathbb{R}^k \rightarrow \mathbb{R}$
 $f(x) \approx f(\bar{x}) + (\nabla_x f(\bar{x}))^T (x - \bar{x})$

At $x = \bar{x}$ $\xrightarrow[\text{first-order approximation}]{f: \mathbb{R} \rightarrow \mathbb{R}}$ $f(x) \approx f(\bar{x}) + \nabla_x f(\bar{x})(x - \bar{x})$



$\nearrow$
 $\searrow$

maximally increase $\rightarrow x - \bar{x} = \eta \nabla_x f(\bar{x})$

maximally decrease $\rightarrow x - \bar{x} = -\eta \nabla_x f(\bar{x})$

step size
 $\eta \geq 0$

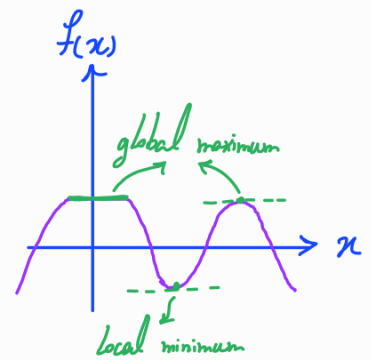
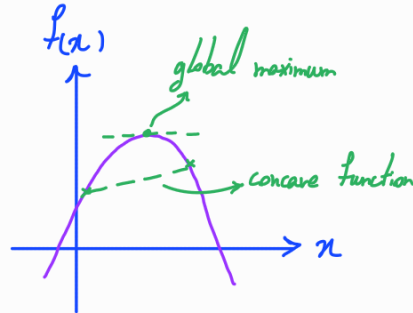
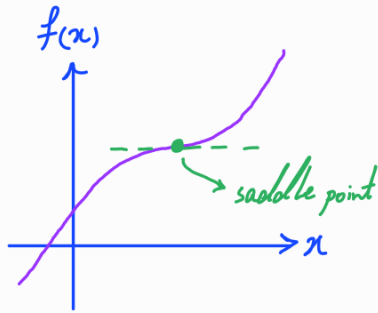
GD update for maximization

$$x_{t+1} = x_t + \eta \nabla_x f(x_t)$$

GD update for minimization

$$x_{t+1} = x_t - \eta \nabla_x f(x_t)$$

Consider a function $f(x) : \mathbb{R}^p \rightarrow \mathbb{R}$ that we want to maximize.



Note: The convergence of first-order optimization techniques such as GD depends on different factors like the function properties, initialization, and the learning rate.

In some settings, it may be infeasible or difficult to compute the gradients, e.g., the function is unknown or the gradient requires computation over all data.

In this case, one can use SGD which works with the gradient estimate.

GD

- Initialize $x = x_0$
- For $t = 0, 1, 2, \dots$
 - Compute $\nabla_x f(x_t)$
 - Update

$$x_{t+1} = x_t - \eta \nabla_x f(x_t)$$

SGD

- Initialize $x = x_0$
- For $t = 0, 1, 2, \dots$
 - Compute g_t s.t. $\mathbb{E}[g_t] = \nabla_x f(x_t)$
 - Update

$$x_{t+1} = x_t - \eta g_t$$

Theorem [Convergence of stochastic Gradient Descent]:

$$\min_{x \in \mathbb{R}^p} f(x)$$

Suppose $f: \mathbb{R}^p \rightarrow \mathbb{R}$ is an objective function with the following properties:

- properties of the objective function
- f is differentiable, i.e., $\nabla_x f(x)$ exists for all x
 - f is β -smooth, i.e.,
$$\|\nabla_x f(x) - \nabla_x f(x_0)\|_2 \leq \beta \|x - x_0\|_2 \text{ for all } x_0 \text{ and } x,$$
 - f is bounded, i.e., $\sup_x |f(x)| \leq M < \infty$.

Running the SGD algorithm with gradient estimates g_t that satisfy

- properties of the gradient estimate
- $\mathbb{E}[g_t] = \nabla_x f(x_t)$, and
 - $\mathbb{E}[\|g_t\|_2^2] \leq G$,

guarantees an asymptotic convergence to a stationary point. In particular,

$$\mathbb{E}\left[\frac{1}{T} \sum_{t=0}^T \|\nabla_x f(x_t)\|_2^2\right] \leq O\left(\sqrt{\frac{\beta M G}{T}}\right),$$

where T is the number of iterations that SGD is run if the step

size η is set to $\eta = \sqrt{\frac{M}{\beta G T}}$.